

Learning2Reason

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<http://www.fnds.cs.ru.nl/fndswiki/Research/Learning2Reason>

Abstract. In recent years, large corpora of formally expressed knowledge have become available in the fields of formal mathematics, software verification, and real-world ontologies. The Learning2Reason project aims to develop novel machine learning methods for computer-assisted reasoning on such corpora. Our global research goals are to provide good methods for selecting relevant knowledge from large formal knowledge bases, and to combine them with automated reasoning methods.

Introduction

In recent years, large corpora of formally expressed knowledge have become available in the fields of formal mathematics, software verification, and real-world ontologies. Examples of such formal mathematical libraries are the Mizar Mathematical Library¹, the Constructive Coq Repository at Nijmegen², the Flyspeck project³, the The Archive of Formal Proofs⁴, and the Mathematical Components project⁵. Real-world ontologies include for example the Cyc Knowledge Base⁶, the Suggested Upper Merged Ontology (SUMO)⁷, and the YAGO semantic knowledge base⁸. To derive new knowledge from them, one uses computer-assisted and automated reasoning methods, which typically use one-problem-at-a-time approaches (e.g., resolution and tableaux proving). This *symbolic* (or *deductive*) approach typically does not consider the knowledge contained in previous proofs, and it suffers from a fast-growing search space.

This makes computer-assisted reasoning a suitable domain for complementary *heuristic* (or *inductive*) AI approaches, like machine learning, which rely on

* We acknowledge support from the Netherlands Organization for Scientific Research, in particular Learning2Reason and a Vici grant (639.023.604).

¹ <http://www.mizar.org>

² <http://c-corn.cs.ru.nl/>

³ <http://code.google.com/p/flyspeck/>

⁴ <http://afp.sourceforge.net/>

⁵ <http://www.msr-inria.inria.fr/Projects/math-components>

⁶ http://cyc.com/cyc/cycrandd/technology/whatisunc_dir/whatsuncyc

⁷ <http://www.ontologyportal.org/>

⁸ <http://www.mpi-inf.mpg.de/yago-naga/yago/>

learning from the large amount of previous knowledge to heuristically control the search space, e.g., by estimating the usefulness of existing lemmas for proving a new result. We propose to use state-of-the-art machine learning techniques, especially kernel-based methods that are ideally suited to take into account the structure of formulas and proofs (given as graphs or trees).

Project Goals

The **Learning2Reason** project aims to propose a theoretical framework and to develop novel machine learning algorithms suitable for formal, computer-assisted, and automated reasoning in the presence of large amount of previous knowledge. The main advantage of our approach is its ability to *learn* and *predict* structural dependencies between theorems and axioms.

For ordinary users of formal systems, software developed in **Learning2Reason** will provide improved mathematical advice using *learned* dependencies mined from the large knowledge bases. For developers of automated reasoning methods, the **Learning2Reason** project will provide learning algorithms that help the process of proof discovery and verification.

Tasks

Our focus so far has been on the following tasks:

1. Developing suitable (structural, semantic) kernels for classification in the formal mathematical domain [3].
2. Formulating the learning task as an instance of different paradigms (binary or multi-class classification, (multi-output) ranking).
3. Developing multi-output rankers for the domain.
4. Defining and creating suitable representations of the mathematical data, and obtaining precise datasets and benchmarks for learning from mathematical libraries.
5. Defining proper machine learning evaluation metrics.
6. Plugging the newly developed machine learning tools into reasoning meta-systems like the Machine Learner for Automated Reasoning (MaLAREa) [4].
7. Evaluation of the combined learning/deductive systems on standard large-theory benchmarks like the MPTP Challenge⁹.

First Results

We conducted first experiments on a subsets of the Mizar mathematical library¹⁰. We compared a regression based kernel with two other ranking algorithms: SNoW [1] in naïve Bayes mode and APRILS, a ranking method which is

⁹ <http://www.tptp.org/MPTPChallenge/>

¹⁰ Available at <http://www.mizar.org>

based on Latent Semantic Analysis [2]. On the test datasets, our kernel algorithm outperforms both methods¹¹.

As an example, we present a comparison of our newly developed MOR algorithm with SNoW's naïve Bayes on the MPTP Challenge¹² problems. We experimentally plugged the MOR algorithm into the MaLAREa system, and compare its speed and precision with MaLAREa running with naïve bayes (SNoW) as a learning algorithm. Only eight MaLAREa iterations are run, in order to remove the gradual effect of using many different specifications, which can with sufficient time equalize any advice algorithm with a random one. The comparison is given in figure 1.

Run	Time Limit	Axiom Limit	Solved Total MOR	Solved Total SNoW
1	1 sec	0	57	58
2	1 sec	256	79	74
3	1 sec	256	83	74
4	1 sec	4	90	84
5	1 sec	8	117	99
6	1 sec	16	135	119
7	1 sec	32	140	127
8	1 sec	64	141	129

Fig. 1. Comparison between MaLAREa-MOR and MaLAREa-SNoW

Even though the MOR algorithm started with one problem less solved, it converges faster and solved 141 problems after eight iterations. With the same number of iterations, MaLAREa with SNoW's naïve Bayes solved only 129 problems. The number of problems solved by the system after the sixth fast one-second run using the MOR-based premise selection outperforms any other non-learning system run in 21 hours on the MPTP Challenge problems. Further information and results can be found on our website <http://www.fnds.cs.ru.nl/fndswiki/Research/Learning2Reason>.

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4. Urban, J., Sutcliffe, G., Pudlák, P.: Malarea SG1-machine learner for automated reasoning with semantic guidance. In: Armando, A., Baumgartner, P., Dowek, G. (eds.) *IJCAR 2008*. LNCS (LNAI), vol. 5195, pp. 441–456. Springer, Heidelberg (2008)

¹¹ to be published, also see <http://www.cs.ru.nl/~kuehlwein/pics/mptp.pdf> and <http://www.cs.ru.nl/~kuehlwein/pics/530.pdf>

¹² <http://www.cs.miami.edu/~tptp/MPTPChallenge/>